

Chanuk Lee

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RESEARCH INTERESTS

I study how to build intelligent systems that make effective use of limited resources: improving reasoning through post-training and enabling models to recognize when their own capabilities are insufficient and selectively use external resources at inference time. My interests include LLM reasoning, post-training, test-time scaling, tool-augmented reasoning, and self-improving agents that learn through interaction.

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST) Mar. 2022 – Present
B.S. in Computer Science; Minor in Mathematics Expected graduation: 2028
Mandatory military service: Republic of Korea Air Force, 2024–2025.

Hansung Science High School Mar. 2020 – Feb. 2022
Early graduation in two years.

RESEARCH EXPERIENCE

KAIST MLAI Lab Oct. 2025 – Present
Undergraduate Research Intern Advisor: Prof. Sung Ju Hwang
Research on LLM reasoning and post-training.

PUBLICATIONS AND PREPRINTS

- [4] **Knowing When Thinking Is Not Enough: Teaching Small Reasoning Models to Reason Beyond Their Parametric Knowledge**
Chanuk Lee, Minki Kang, Sangwoo Park, Woongyeong Yeo, Jinheon Baek[†], Sung Ju Hwang[†]
arXiv preprint, 2026. [Paper] [Code]
- [3] **Surprising Success, Repeated Failure: Entropy-Guided Credit Assignment for Exploration in LLM Reasoning**
Woongyeong Yeo, Minki Kang, Chanuk Lee, Sangwoo Park, Jinheon Baek[†], Sung Ju Hwang[†]
arXiv preprint, 2026. [Paper] [Code]
- [2] **Nudging Beyond the Comfort Zone: Efficient Strategy-Guided Exploration for RLVR**
Chanuk Lee^{*}, Sangwoo Park^{*}, Minki Kang, Sung Ju Hwang
MATH-AI@NeurIPS 2026 Workshop. [Paper] [Code]
- [1] **SAGE: Shaping Anchors for Guided Exploration in RLVR of LLMs**
Chanuk Lee, Minki Kang, Sung Ju Hwang
International Conference on Machine Learning (ICML), 2026. [Paper] [Code]

^{*} Equal contribution [†] Equal advising / Corresponding authors

ACADEMIC SERVICE

Reviewer: MATH-AI@NeurIPS 2026 Workshop.

TECHNICAL SKILLS

Programming	Python
ML/LLM	PyTorch, Transformers, PEFT, TRL, verl, vLLM
Tools	Weights & Biases, Git, Linux, L ^A T _E X
Languages	Korean, English